Modding

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Modding

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Category:Modding • Scripts • Lua (API) • Java • User Interface • Modeling • Animation • Texturing • Mapping • Rendering • Translation

This page has been updated to an *unstable* beta version (42.14.1).

This article is about an overview about modding. For PZwiki project, see PZwiki:Project Modding.

Modding is the process of creating new content by creating mods for a game. For Project Zomboid, modding is a natively integrated part of the game with a mod manager and an API to allow modders to create a various quantity of content, from simple items to complex systems. Mods are usually uploaded to the Spiffo's Workshop which serves as the official method for sharing mods with the community.

Spiffo's Workshop is the home of Project Zomboid mods on Steam

Latest information on MP unstable release

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How to get started

You can learn more in detail how to get started with modding in the getting started with modding page, which will also teach you about various tools for different modding fields. Depending on the type of modding you are interested in, the #Modding Resource Hub will provide you with the necessary resources to get started as well as links to some important pages per modding fields.

Modding fields

Modding is split in different fields, each with their own tools and creation process. For a few general resources on modding that you will need to know regardless of the field you are working on:

- Game files - Explains Project Zomboid's file structure.
- File formats - Documentation of file formats used by the game.
- Mod structure - Explains a mod files structure.
- Debug mode - Explanations of how to run the game in the debug (developer) mode and how to use it.
- Startup parameters - Startup parameters to launch the game with.
- Mod optimization - Learn various tips, tricks and good habits that will improve your mods performances.
- Procedural distributions - List and explanations of procedural distributions.

Below is a list of all the modding fields with a brief description of what they involve and links to the relevant pages:

Modding fields

Field	Description	Important pages
Scripts (Category:Scripts)	Allow to add items in the game with simple parameters which can be edited in any text editor. No programming knowledge is needed.	
Lua API (Category:Lua (API))	Allows modders to program functionalities via a Java implementation of Lua called Kahlua which enables Lua scripts to run within Java programs. It bases itself on Lua 5.1 (https://www.lua.org/manual/5.1/) with a few differences and allows the use of exposed Java class and methods from Lua scripts.	• Lua (language) to learn how to code with the Lua programming language. • Lua object – a full list of available Lua Objects to be used by modders. • Lua event – a full list of available Events to hook on. • Java object – a list of informations about some Java objects which can be used with the Lua API. • JavaDocs - a list of external links to the Java documentation of game. • Decompiling game code - a guide to decompiling the Java to better understand how the game works.
Java (Category:Java)	Involves modifying the Java game code directly and the manual installation of the Java files to modify deep game functionalities. It is not recommended for beginners as it requires a good knowledge of Java and programming in general.	
Modeling	Creating 3D models to be used in item making, vehicle making or renders.	• Creating custom animations • Creating a hair mod • Creating a clothing mod

Field	Description	Important pages
Texturing	Creating textures to be applied on 3D models or to create 2D tiles.	
Rendering	Creating 2D images from 3D models to be used in various medias in-game such as fliers.	
Animation (Category:Animation)	Used to create new animations for characters to visually do certain actions. It involves knowledge of 3D softwares for animating but also a small bit of XML programming required to get your animation in-game.	
Mapping (Category:Mapping)	Create new locations, buildings and general environments in the form of a map for users to play on.	• map.info - Definition of the file used to define the map informations. • Tiledefs used by mods - List of tiledef IDs which are already used by other mods. • Adding new tiles - Guide to adding new tiles. • Tile properties - Explanation of tile properties. • Room definitions and item spawns - Explanation of room definitions, used mainly for loot distribution. • Vehicle zones - Explanation of vehicle zones.
Translation	Translating the game or mods to various languages.	

Modding guides

The wiki provides various modding guides as well as individual tutorials for different modding fields. You can find the full list of guides in the Modding guides category. Each individual guide pages will also link to external guides on the same subject.

Unlisted external guides are listed below:

External tutorials

Tutorial	Type	Description	Author	Last updated
PZ Modding Video Guides (https://www.youtube.com/playlist?list=PL91PT-Vd7Vhp_eBA8pE4hCXPcrxHWgr_E)	General guide	Video guides for modding Project Zomboid, covering various aspects of modding.	SimKDT	2025 November 19
PZ Modding Guides (https://github.com/demiurgeQuantified/PZModdingGuides)	General guide	A collection of guides and documentation for modding Project Zomboid.	Albion	2025 May 18
Project Zomboid B42 - How to Create a Mod (https://www.youtube.com/watch?v=yFMLmgM20xc)	Item creation	A short video tutorial explaining how to create an item mod for Project Zomboid.	Rainmaker	2025 April 23
How to create Custom Firearm with working muzzle/attachments (https://steamcommunity.com/sharedfiles/filedetails/?id=3060255898)	Item creation	A guide on how to create a custom firearm with working muzzle and attachments.	Nik	2024 June 29
How to change Zomboid's in-game music and sound effects (https://steamcommunity.com/sharedfiles/filedetails/?id=2474838710)	Sound	A guide on how to change the in-game music and sound effects of Project Zomboid.	Sokolov	2024 April 2
PZ Mod Documentation (https://github.com/MrBounty/PZ-Mod---Doc)	General guide	A collection of guides and documentation for modding Project Zomboid.	MrBounty	2023 August 25
FWolfe's Modding Guide (https://github.com/FWolfe/Zomboid-Modding-Guide) (WIP)	General guide	A comprehensive guide to modding Project Zomboid, covering various aspects of modding. However a lot of the information is outdated or better explained by the official wiki. Good pratices are notably way more up-to-date than the wiki. The guide is incomplete on a lot of aspects and now inactive.	Fenris Wolf	2023 January 22
Full Project Zomboid Mod Tutorial - Start to Finish (https://www.youtube.com/watch?v=-yrmCAwzTbY)	Item creation	A 2 hour and a half long step-by-step video tutorial on how to add a new item to Project Zomboid, including creating an icon, adding it to the game, testing it and uploading it to the Steam Workshop.	W. Patrick	2022 December 13

External resources

External resources

Resource	Description	Author	Last updated
Mod Resources (Official Discord - Thread The Under Mod Development Channel) (https://discord.com/channels/136501320340209664/1070852229654917180)	A thread on the official Discord server that contains a list of resources for modding Project Zomboid, including links to resources, tools, and other useful information.	Glytch3r and more	2025 July 4

General additional modding resources

- Photoshop Masks used for making and editing tile sprites, a link to a post on TIS Discord with the file (https://discord.com/channels/136501320340209664/279303619277488128/864229655867031572)
- A flow chart for the timed actions (https://github.com/Poltergeistzx/Project-Zomboid/blob/main/docs/TimedActions%20Flow%20Chart.pdf)
- SpawnerAPI: Allows for pending the spawns of vehicles, items, zombies in order to spawn things anywhere in the world. Upon loading the cell in question the item becomes spawned in (https://github.com/Konijima/PZ-Community-API/tree/master/Contents/mods/CommunityAPI/media/lua/client/SpawnerAPI)
- How to spawn loot on specific zombie outfit corpses: a link to a post on TIS Discord explaining how (https://discord.com/channels/136501320340209664/232196827577974784/891769429543759883)
- WordZed tutorial on YouTube (https://www.youtube.com/watch?v=iv7N7gmowWU)
- Complete Vehicle Modding Tutorial on TIS forum (https://theindiestone.com/forums/index.php?/topic/28633-complete-vehicle-modding-tutorial/)
- isoRangeScan: This is a utility function meant for large-scale scans of isoGridSquares around a given IsoObject. The scans are done fractally – that is to say, from a center (or centers) outward to fill a larger area. (https://github.com/Konijima/PZ-Community-API/blob/master/Contents/mods/CommunityAPI/media/lua/shared/CommunityAPI/IsoUtils.lua)

Navigation

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